

Software-reconfigurable wavelength-tunable dual optical frequency comb based on externally-injected gain-switched lasers

Minghao Wei⁽¹⁾⁽²⁾, Alejandro Rosado⁽¹⁾⁽²⁾, Conor McArdle⁽¹⁾, and Prince M. Anandarajah⁽¹⁾⁽²⁾

⁽¹⁾Photonic Systems and Sensing Laboratory (PSSL), School of Electronic Engineering, Dublin City University, Ireland, alejandro.rosado@dcu.ie

⁽²⁾CONNECT Research Centre, School of Engineering, Trinity College Dublin, Ireland.

Abstract We demonstrate a software-reconfigurable wavelength-tunable dual optical frequency comb based on externally injected gain-switched lasers, achieving >30 nm contiguous tunability with minimum step of 2 pm. The system operates without wavelength-dependent re-optimisation while maintaining consistent DOFC performance across the tuning range, enabling broadband sensing.

Introduction

DOFC-based systems have been widely applied in areas such as gas sensing [1], fibre Bragg grating interrogation [2], and distance ranging [3], owing to their single-shot measurement capability and simplified signal acquisition. Both these features are enabled by the dual-comb architecture, in which two optical frequency combs (OFCs) with slightly different free spectral ranges (FSRs) generate a radio-frequency (RF) comb through multi-heterodyne detection. Such a direct translation to lower frequencies enables the effective mapping of the optical spectral information into the RF domain [1]. Various DOFC generation techniques have been reported, including mode-locked laser (MLL)-based [4], electro-optic modulator (EOM)-based approaches [5], and externally injected gain-switched laser (EI-GSL)-based schemes [6]. Among these, the EI-GSL-based DOFC generation provides a simple and cost-effective architecture, which is amenable to be realised as a photonic integrated circuit (PIC) [7]. In addition, it also offers flexible wavelength and FSR control, all of which make it attractive for practical sensing applications [8].

Despite these advantages, EI-GSL-based DOFCs typically exhibit a relatively narrow optical bandwidth, fundamentally limited by the frequency response of the semiconductor laser. The narrower spectral span of these combs limit their employment in applications requiring wide spectral coverage. Extensive efforts have therefore been devoted to overcoming this limitation through various comb expansion techniques [9]. Nevertheless, it remains challenging to directly generate wideband comb spectra spanning several nm, without relying on additional complexity and components exhibiting nonlinearity. Hence, it could be more cost and power efficient to exploit the inherent flexibility offered by the EI-GSL-based OFC generation schemes, specifically in terms of wavelength tunability [10].

Previous reports [11, 12] of wavelength tunability in EI-GSL based schemes have typically been limited to discrete operating points [11] or restricted tuning ranges of few nm [12]. In addition, the comb at different central emission wavelength exhibited large variations in the spectral amplitude/flatness across the tuning range. Typically, in order to maintain a spectrally flat, high-quality comb requires the optimisation of wavelength-dependent parameters such as the modulation amplitude and DC bias current [11]. Hence, a major unaddressed gap in EI-GSL based wavelength tunable DOFCs is the systematic demonstration of contiguous tuning and uniformity of the spectral flatness.

In this work, we propose for the first time a contiguously wavelength-tunable EI-GSL-based DOFC, achieving a tuning range exceeding 30 nm with a minimum tuning step of 2 pm. Wavelength tuning is realised by directly controlling the wavelength of the primary-laser and the temperatures of the secondary lasers. This enables software-defined, automated, and repeatable operation while maintaining consistent comb quality across the tuning range, significantly extending the operational capability of EI-GSL-based DOFCs. To the best of our knowledge, this is the first demonstration of a software-defined contiguously tunable DOFC.

System Setup and Tuning Mechanism

The proposed experimental setup for the generation and characterisation of the wavelength tunable EI-GSL-based DOFC is illustrated in Fig. 1. The system adopts a primary–secondary laser configuration, consisting of a narrow-linewidth semiconductor tunable laser (TL) acting as the common primary laser (PL) and two gain-switched (GS) Fabry–Perot (FP) lasers as secondary sources. Each FP is modulated by an amplified sinusoidal RF signal, as indicated by the dashed blue box in Fig. 1. Under optical injection, a selected longitudinal mode of each FP laser is injection-locked, while the remaining modes are

suppressed [13]. By gain-switching the FP lasers with RF signals at slightly different modulation frequencies, two OFCs with a small FSR offset are generated. Since both OFCs are optically injection-locked by the same PL, one comb line, from each OFC is naturally aligned at the injection wavelength, providing a common optical reference and ensuring mutual coherence. The combined OFCs are fed to a low speed photodetector, to down-convert the optical combs to an RF comb, which enables the mapping of the optical spectral information into the RF domain. As previously mentioned, one of the shortcomings of the EI-GSL technique is the limited 3 dB bandwidth of the generated comb (0.4 – 0.5 nm).

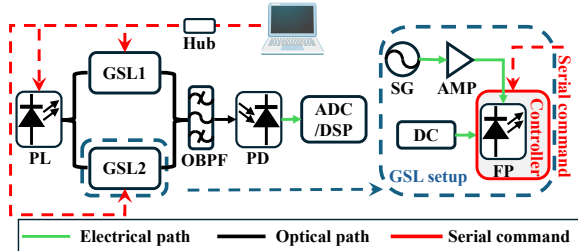


Fig. 1: Experimental setup of software-tunable EI-GSL-based DOFC source., PL (primary laser), GSL (gain-switched laser), PD (photodetector), SG (RF signal generator), AMP (RF amplifier), ADC (analog-to-digital converter).

To overcome this limitation, wavelength tuning is targeted and achieved by exploiting the longitudinal-mode structure of the FP lasers. For each FP laser, different longitudinal modes can be selectively injection-locked, enabling DOFC generation at discrete wavelengths. Besides, by adjusting the operating temperature, the longitudinal modes shift due to variations in the effective refractive index and cavity length, allowing a precise alignment between the selected mode and the injected wavelength.

The proposed tuning scheme starts from a reference DOFC operating point, where all system parameters are optimised. Rather than re-optimising these parameters for each target wavelength, wavelength tuning is performed by updating only the PL wavelength and FP laser temperatures. The tuning procedure can be summarised as follows:

1. A target injection wavelength λ_{target} is defined, and the PL is tuned to this wavelength.
2. The nearest longitudinal mode λ_{NearMod} is identified based on the pre-characterised FP laser's spectra.
3. The wavelength offset is calculated as

$$\Delta\lambda = \lambda_{\text{target}} - \lambda_{\text{NearMod}}. \quad (1)$$

4. Based on the approximately linear dependence between wavelength and temperature, the required temperature shift

ΔT is obtained as

$$\Delta T = \frac{\Delta\lambda}{0.104}, \quad (2)$$

where the calibrated coefficient $\Delta\lambda/\Delta T \approx 0.104 \text{ nm}/^\circ\text{C}$ arises from the temperature-dependent variation of the effective refractive index and cavity length [14].

5. The β -model is then used to determine the corresponding controller setpoints:

$$\frac{1}{T} = \frac{1}{T_0} + \frac{1}{\beta} \ln\left(\frac{R}{R_0}\right), \quad (3)$$

where $\beta = 3930 \text{ K}$ is the thermistor constant provided by the manufacturer, R is the resistance corresponding to the target temperature T , and R_0 is the resistance at the reference temperature T_0 , enabling precise mapping from the target temperature to the controller resistance settings.

Through this process, the nearest longitudinal modes are aligned with the target injection wavelength, allowing wavelength tuning without modifying the RF drive amplitude or DC bias current. This reduces the tuning process from a wavelength-dependent multi-parameter optimisation problem to a direct wavelength-to-temperature mapping, enabling automated DOFC operation with a reduced parameter space.

Experimental Results

The wavelength tunability of the proposed DOFC is experimentally evaluated in the optical domain across a range from 1535 nm to 1565 nm, where 33 operating points are characterised. All generated OFCs exhibit 3 dB spectral widths exceeding 0.4 nm. Among these, 12 representative operating points are selected, and the corresponding optical spectra of OFC1 and OFC2 are shown separately (for clarity) in Fig. 2(a) and Fig. 2(b). To support the contiguous tuning capability, an exemplar wavelength span of 10 nm (1545 to 1555 nm) is selected (24 out of the 33 operating points), with the target wavelength incremented in steps of 0.4 nm. Since the 3 dB spectral width of each OFC exceeds the step size, adjacent comb spectra overlap, enabling gap-free spectral coverage. In addition, fine wavelength tuning with a minimum step of 2 pm is achieved, which is limited by the resolution of the tunable laser used in the experiment.

To further quantify the comb performance across the tuning range, a figure of merit (FoM) [15] is employed, which is defined as the product of the comb-line carrier-to-noise ratio (CNR) and the 10-dB spectral width Δf_{10} . The CNR is

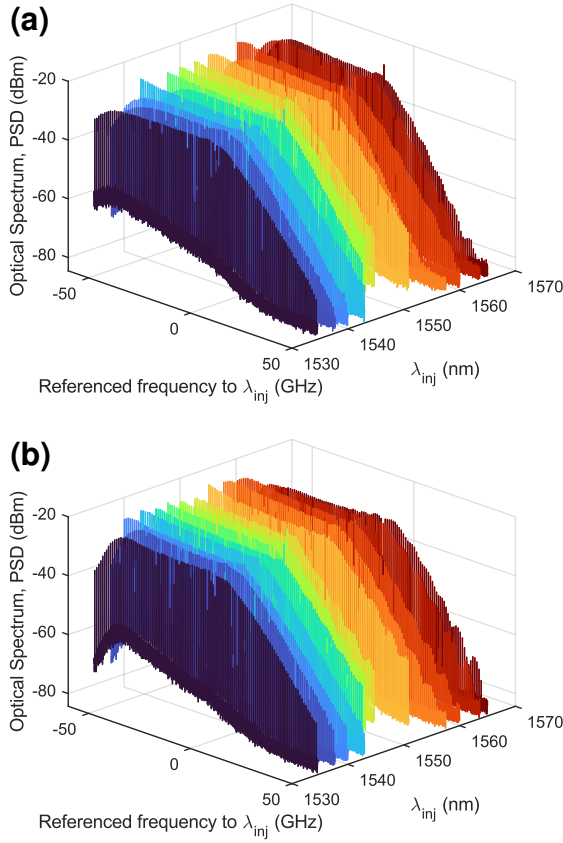


Fig. 2: Example optical spectra at different target wavelengths: (a) OFC1, and (b) OFC2.

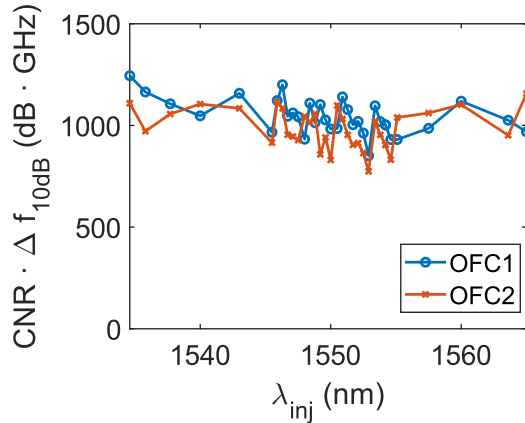


Fig. 3: Figure of merit of the OFCs at different target wavelengths.

calculated as the average signal-to-noise ratio of the comb lines within 10 dB from the peak, and Δf_{10} represents the maximum spectral span within the same 10 dB envelope. This metric therefore captures both the spectral quality and usable bandwidth of the comb. The results, shown in Fig. 3, indicate that both OFC1 and OFC2 maintain comparable FoM values across the full 30 nm range, confirming that the comb quality remains uniform over the entire tuning span. The DOFC tuned to each wavelength is subsequently heterodyne detected to generate an RF comb. Representative RF comb spectra obtained at different target wavelengths across the full tuning

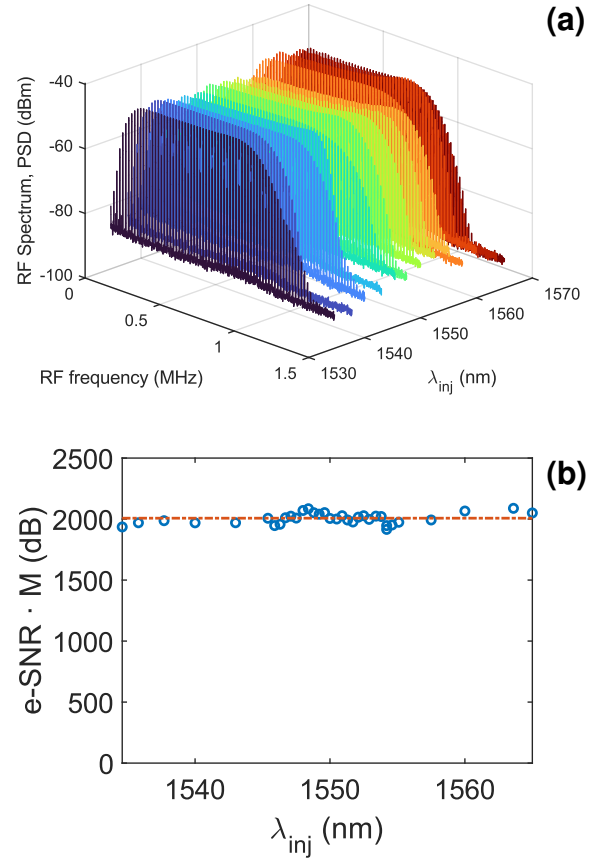


Fig. 4: Characteristics of the RF comb at different target wavelengths: (a) sample of RF comb spectra, (b) figure of merit of the RF comb.

range are shown in Fig. 4(a), demonstrating that the corresponding RF combs retain similar spectral characteristics at each of the chosen operating points. The quality of the RF comb is also evaluated using an electrical figure of merit (eFoM), defined as the product of the electrical signal-to-noise ratio (e-SNR) and the number of tones (M) within the 10-dB window, as shown in Fig. 4(b). The results indicate comparable eFoM values across the entire tuning range, confirming consistent RF comb performance.

Conclusion

We demonstrate a wavelength-tunable EI-GSL-based DOFC achieving >30 nm tuning range with a minimum step of 2 pm without wavelength-dependent re-optimisation. By controlling only the primary-laser wavelength and FP laser temperatures, automated tuning is achieved with a reduced parameter set. Similar FoM (~ 1000 dB · GHz) and eFoM (~ 2000 dB) across the tuning range indicate consistent DOFC quality in both optical and RF domains. These results confirm a robust and scalable wavelength-agile dual-comb platform, extending the capability of EI-GSL-based DOFCs for wide spectral coverage and high-resolution sensing, such as multiplexed FBGs and tilted FBG sensing.

Acknowledgements

This work was supported by Taighde Éireann – Research Ireland under Grant No. 13/RC/2077_P2 at CONNECT: the Research Ireland Centre for Future Networks.

References

- [1] I. Coddington, W. C. Swann, and N. R. Newbury, "Coherent multiheterodyne spectroscopy using stabilized optical frequency combs", *Physical Review Letters*, vol. 100, no. 1, p. 013902, 2008. DOI: 10.1103/PhysRevLett.100.013902.
- [2] R. Zhang, Z. Zhu, and G. Wu, "Static pure strain sensing using dual-comb spectroscopy with fbg sensors", *Optics Express*, vol. 27, no. 23, pp. 34 269–34 283, 2019. DOI: 10.1364/OE.27.034269.
- [3] P. Trocha et al., "Ultrafast optical ranging using microresonator soliton frequency combs", *Science*, vol. 359, no. 6378, pp. 887–891, 2018. DOI: 10.1126/science.aao3924.
- [4] S. M. Link, D. Maas, D. Waldburger, and U. Keller, "Dual-comb spectroscopy of water vapor with a free-running semiconductor disk laser", *Science*, vol. 356, no. 6343, pp. 1164–1168, 2017. DOI: 10.1126/science.aam7424.
- [5] M. Yan, P.-L. Luo, K. Iwakuni, G. Millot, T. W. Hänsch, and N. Picqué, "Mid-infrared dual-comb spectroscopy with electro-optic modulators", *Light: Science & Applications*, vol. 6, no. 10, e17076–e17076, 2017. DOI: 10.1038/lsa.2017.76.
- [6] B. Jerez, P. Martín-Mateos, E. Prior, C. de Dios, and P. Acedo, "Dual optical frequency comb architecture with capabilities from visible to mid-infrared", *Optics Express*, vol. 24, no. 13, pp. 14 986–14 994, 2016. DOI: 10.1364/OE.24.014986.
- [7] A. Rosado, M. D. Gutierrez-Pascual, G. Jain, F. Smyth, and P. M. Anandarajah, "Dual optical frequency comb using a multi-functional pic based on fully integrated injection locked gain-switched lasers", *Scientific Reports*, 2025. DOI: 10.1038/s41598-025-33112-5.
- [8] E. P. Martin, S. T. Ahmad, S. Chandran, A. Rosado, A. A. Ruth, and P. M. Anandarajah, "Stability characterisation and application of mutually injection locked gain switched optical frequency combs for dual comb spectroscopy", *Journal of Lightwave Technology*, vol. 41, no. 13, pp. 4516–4521, 2023. DOI: 10.1109/JLT.2023.3255550.
- [9] P. D. Lakshmijayasimha, A. Kaszubowska-Anandarajah, E. P. Martin, P. Landais, and P. M. Anandarajah, "Expansion and phase correlation of a wavelength tunable gain-switched optical frequency comb", *Optics Express*, vol. 27, no. 12, pp. 16 560–16 570, 2019. DOI: 10.1364/OE.27.016560.
- [10] M. D. G. Pascual, R. Zhou, F. Smyth, P. M. Anandarajah, and L. P. Barry, "Software reconfigurable highly flexible gain switched optical frequency comb source", *Optics Express*, vol. 23, no. 18, pp. 23 225–23 235, 2015.
- [11] L. Monroy, A. Pérez-Serrano, J. M. G. Tijero, and I. Esquivias, "Multi-Gas Dual-Comb Spectroscopy With Tunable Gain-Switched Laser Diodes", *Scientific Reports*, vol. 15, no. 1, p. 7610, 2025. DOI: 10.1038/s41598-025-7610X.
- [12] E. Russell, A. A. Ruth, B. Corbett, and F. C. G. Gunning, "Tunable Dual Optical Frequency Comb at 2 μ m for CO₂ Sensing", *Optics Express*, vol. 31, no. 4, pp. 6304–6313, 2023. DOI: 10.1364/OE.479607.
- [13] P. Anandarajah, P. J. Maguire, A. Clarke, and L. P. Barry, "Self-seeding of a gain-switched integrated dual-laser source for the generation of highly wavelength-tunable picosecond optical pulses", *IEEE Photonics Technology Letters*, vol. 16, no. 2, pp. 629–631, 2004. DOI: 10.1109/LPT.2003.821252.
- [14] J. O'Carroll, R. Phelan, B. Kelly, D. Byrne, L. P. Barry, and J. O'Gorman, "Wide temperature range 0< t< 85 c narrow linewidth discrete mode laser diodes for coherent communications applications", *Optics Express*, vol. 19, no. 26, B90–B95, 2011. DOI: 10.1364/OE.19.000B90.
- [15] A. Rosado, A. Pérez-Serrano, J. M. G. Tijero, A. V. Gutierrez, L. Pesquera, and I. Esquivias, "Numerical and experimental analysis of optical frequency comb generation in gain-switched semiconductor lasers", *IEEE Journal of Quantum Electronics*, vol. 55, no. 6, pp. 1–12, 2019. DOI: 10.1109/JQE.2019.2943482.